DOU, HONGWEN *Ph.D.*

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EDUCATION

Ph.D. in Building Engineering 2019-2023

Concordia University, Canada

Master of Applied Science in Architecture and Civil Engineering 2016-2018

Wuhan University of Technology, China

Bachelor's in Building Environment and Equipment Engineering 2010-2014

North China University of Water Resources and Electric Power, China

RESEARCH EXPERIENCES

Postdoctoral Fellowship, ÉTS, Canada 2024-Present

Model predictive control for building energy flexibility

- Engineered deep-learning-driven Model Predictive Control (MPC) frameworks to forecast and manage demand-side energy consumption for the cooling and heating scenarios.
- Implemented a multi-objective MPC strategy in commercial infrastructure to enhance building load shifting capabilities and minimize energy cost while maintaining indoor thermal comfort.
- Performed techno-economic assessments of large-scale residential electric heating flexibility in cold climate zones, leveraging real-world consumption datasets from a local utility company.

The Building Optimization Testing Framework 2 (BOPTEST2)-Task 3: Emulator development to enhance the energy efficiency of school buildings

- Architected and optimized an operational energy flexibility strategy for existing primary schools, focusing on peak-load management and optimization during Quebec's intensive heating seasons.
- Designed and validated a Modelica model for a thermal storage device, implementing advanced control logic for seamless HVAC integration.
- Developing comprehensive building emulators, incorporating multi-zone thermal dynamics and complex HVAC sequences, to benchmark and validate demand-side management strategies.

Detection and diagnosis of multiple-dependent faults in chillers

- Formulated and validated physical-based chiller models by performing thermodynamic analysis and heat transfer analysis.
- Tested and verified the model response to multiple concurrent faults by deploying it to a fault detection framework; designed rule-based fault diagnosis approaches to identify fault root causes.

Research Associate, Concordia University, Canada 2023-2023

Building indoor thermal comfort under summer overheating conditions

- Analyzed field measurement data of indoor and outdoor conditions to estimate building thermal time constants.
- Proposed and optimized forecasting strategies using time-series models to forecast indoor thermal conditions and provide early overheating warnings.

Research Assistant, Concordia University, Canada 2019-2023

Fault detection and diagnosis for building automation systems

- Delivered benchmarking grey-box and deep learning models to estimate the operation characteristics of chillers under both system-level and component-level.

- Developed and validated fault detection and diagnosis models for multiple-dependent chiller faults.
- Investigated the transfer learning approach to enhance the model performance.
- Conducted long-term energy performance analysis for a large chilled water plant to assess its operation efficiency.

Research Assistant, Wuhan University of Technology, China 2016-2019

Zero-energy buildings in the cold-winter and hot-summer climate zone

- Conducted the feasibility study of zero-energy residential houses with on-site PV systems.
- Simulated and optimized the influences of building envelope configurations, PV installation angles, and HVAC control strategies, etc. on net-zero energy performance.

High-efficiency thermal system design

- Developed patented intelligent heat pump system to achieve simultaneous cooling, heating, and hot water production, maximizing COP, and minimizing thermal pollution.

Measurement and simulation of building outdoor environments

- Designed and conducted field measurement campaigns in metropolitan central areas; combined with CFD simulations, analyzed the urban heat island effect and gaseous pollutant dispersion.

INDUSTRIAL EXPERIENCE

Henan Xuantong Energy Technology Co., Ltd., China 2015-2016

Assistant consultant and designer

- Conducted a feasibility study for a large cooling station in the central business district of Zhengzhou.
- Assisted in designing HVAC system (air distribution, hydronic systems, equipment sizing, etc) for a medium-sized office building.

Shanghai Road and Bridge (Group) Co., Ltd., Shanghai, China 2014-2015

Cost engineer and contract administrator

- Cost engineer: Measured and verified the quantities of completed work, and made payments for contractors and subcontractors.
- Contract administrator: Assisted in drafting, reviewing, and auditing construction project contracts.

RESEARCH GRANTS

Novel method to detect and diagnose multiple dependent faults of building mechanical systems

- PI: **Hongwen Dou**, ÉTS.
- Funding agency: Quebec Research Fund – Nature and Technology (FRQNT), $35,000.

TEACHING&MENTORING EXPERIENCE

Teaching assistant at Concordia University 2022-2022

BLDG 365: Building Science Course

- Led lab sessions for 30+ undergraduate students across six technical experiments.
- Delivered five lectures and evaluated student performance by grading assignments and exams.

Mentor at Wuhan University of Technology, China 2017-2018

Wuhan University of Technology, China

- Supervised three undergraduate students in applying for the national innovation training project.
- Mentored them through performing calculations, analyzing results, and preparing the final report.

PUBLICATIONS

A. Peer-reviewed Journal Papers:

1) **H. Dou**, K. Zhang, R. Zmeureanu. Grey-box models of chiller evaporator for practical integration in building automation systems. *Science and Technology for the Built Environment*.2026, (2)32: 263-280.
2) **H. Dou**, K. Zhang. Transfer learning for cross-building forecasting of building energy and indoor air temperature in model predictive control applications. *Journal of Building Engineering*. *2025, 113341*
3) R. Zmeureanu, **H. Dou,** H. Ge, L. Wang, Z. Xie. Thermal time constant estimation of unoccupied school buildings from field measurements over summer. *Journal of Building Engineering*. *2025, 112311*
4) **H. Dou**, R. Zmeureanu. Transfer learning for the prediction of energy performance of water-cooled electric chillers: grey-box models versus deep neural network (DNN) models. *Energies 2024, 17(23), 5981*
5) **H. Dou**, R. Zmeureanu. Deep neural network models supported by transfer learning for the prediction of operation performance of chillers. *Energies 2023, 16(20), 7149*
6) **H. Dou**, R. Zmeureanu. Detection and diagnosis of multiple-dependent faults (MDFDD) of water-cooled centrifugal chillers using grey-box model-based method. *Energies* 2023, 16(1), 210
7) **H. Dou**, R. Zmeureanu. Evidence-based assessment of energy performance of two large centrifugal chillers over nine cooling seasons. *Science and Technology for the Built Environment* 2021:1–19
8) **H. Dou**, T. Ming, Z. Li, C. Peng, C. Zhang, X. Fu. Numerical simulation of pollutant dispersion characteristics in a three-dimensional urban traffic system. *Atmospheric Pollution Research* 2018;9:735–46
9) Z. Li, C. Cai, X. Huang, **H. Dou**, W. Fang, T, Ming. Numerical simulation on the effect of viaduct settings on the airflow and pollutants dispersion in the deep street canyon. *Research of Environmental Sciences* 2018;31(2):254–64
10) **H. Dou**, T. Ming, J. Xu, Z. Li, C. Cai, W. Fang. Numerical study of gaseous pollutant propagation in a tri-dimensional urban transportation systems. *China Environmental Science* 2018; 38(1): 51-58
11) Y. Li, **H. Dou**, Y. Lin. Analysis on Zero Energy Building in Wuhan. *Building Energy Efficiency*. 2017;45:28–31

B. Peer-reviewed Conference Papers:

1) Y. Chai, **H. Dou**, K. Zhang. Assessing the operational performance and financial incentive of a large-scale demand response program. eSim2026, Montréal, Canada (under review)
2) **H. Dou**, K. Zhang. Development and Validation of a Generic Evaporator Model of Chillers. IBPSA-World conference 2025, August 24-27, 20254, Brisbane, Australia
3) **H. Dou**, K. Zhang. Transfer learning-supported building electric energy forecasting using neural network models. IBPSA-Asia ASim conference 2024, December 8-10, 2024, Osaka, Japan
4) **H. Dou**, R. Zmeureanu. Grey-box models supported by transfer learning for improving the prediction of ongoing commissioning of water-cooled electric chillers. IBPSA Canada eSim conference 2024, June 5-7, 2024, Edmonton, Canada

PATENTS

1) L. Zhang, W. Yang, **H. Dou**, J. Z., S. Xia, Y. Zhang., K. Wu. (2021). A series connection system to recover the heat from the heat pump condenser. (C.N. Patent No. CN 212566374 U). China National Intellectual Property Administration
2) L. Zhang., W. Yang, **H. Dou**, S. Xia, J. Zhang, Y. Zhang, K. Wu. (2021). A condensing water recycling system for centralized HVAC. (C.N. Patent No. CN 212566239 U). China National Intellectual Property Administration
3) T. Ming, D. Li, C. Cai, W. Fang, **H. Dou**, T. Gan, Z. Zhao. (2018). An innovative water-source air conditioner with a cooling tower (C.N. Patent No. CN 207922450 U). China National Intellectual Property Administration
4) **H. Dou**, W. Yang. (2017). A domestic water heater integrated with the split air-conditioner (C.N. Patent No. CN 206398896 U). China National Intellectual Property Administration
5) **H. Dou**, T. Ming, C. Cai, W. Fang. (2017). An improved direct expansion system of air source heat pump to supply space cooling, space heating, and domestic hot water (C.N. Patent No. CN 206310570 U). China National Intellectual Property Administration

PRESENTATIONS

1) Development and Validation of a Generic Evaporator Model of Chillers. BS2025, the 19th IBPSA-World Conference, Brisbane, Australia, August 24-27, 2025
2) Transfer learning-supported building electric energy forecasting using neural network models. ASim 2024, the 5th Asia Conference of the IBPSA, Osaka, Japan, December 8-10, 2024
3) Grey-box models supported by transfer learning for improving the prediction of ongoing commissioning of water-cooled electric chillers. eSim 2024, the 13th Conference of IBPSA-Canada, Edmonton, Canada, June 5-7, 2024

SERVICE&OUTREACH

Conference Session Chairs

- The *19th IBPSA-World Conference, Brisbane, Australia* 2025
- The *5th Asia Conference of the IBPSA: Asim 2024, Osaka, Japan* 2024

Paper Reviewer

- Renewable and Sustainable Energy Reviews
- The Journal of Supercomputing
- Energy Efficiency
- Buildings
- Energies
- Applied Sciences
- Processes
- Proceedings of the 18th IBPSA-World Conference
- Proceedings of the *5th* IBPSA-Asia Conference
- Proceedings of the 19th IBPSA-World Conference

Membership

- E.I.T, APEGS Since 2025
- International Building Performance Simulation Association (IBPSA) World and Canada Since 2020

SKILLS

Programming: Python, MATLAB, C
Energy simulation: Modelica, EnergyPlus, eQUEST, OpenStudio, Autodesk Ecotect
CFD: Ansys Fluent
Optimizer&Analysis: Scipy, Pandas, scikit-learn

SELECTED AWARDS & HONORS

Merit Scholarship Program for Foreign Students, FRQNT, Quebec Government	2025-2026
Concordia International Tuition Award of Excellence, Concordia University	2019 - 2023
First-class Academic Scholarship (twice), Wuhan University of Technology	2017 - 2018
National Scholarship for Graduate Students (top 0.2%), China	2017
Outstanding employee, Shanghai Road and Bridge (Group) Co., Ltd.	2015 - 2016
The Third Prize of National HVAC Science and Technology Competition, China	2014
The First Prize of Provincial HVAC Science and Technology Competition, China	2014
Outstanding Student Scholarship (three times), North China University of Water Resources and Electric Power	2012 - 2014